Electric Vehicle Management System Using Artificial Intelligence and Machine Learning

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

There is a growing need for advanced post-sale management solutions for electric vehicles (EVs) that extend beyond just monitoring their status and performing reactive maintenance. Current EV management systems are designed mainly to track the current state of an EV (in real-time) and to carry out manual diagnostics of that EV, thereby providing limited capabilities for providing intelligence about long-term performance, detecting early failures, and providing data-driven decision support. This demonstrates a significant gap in research to utilize operational data post-sale to make proactive predictions of failures and optimize EV performance throughout the entire lifecycle of the vehicle.

The Post Sale Electric Vehicle Performance Intelligence Failure Prediction System (PSEVPIFPS) represents a new element of an Electric Vehicle Management Systems (EVMS) that utilizes Artificial Intelligence (AI) and Machine Learning (ML) technologies to analyze battery conditions, driving behavior's, ambient conditions, and past sensor data continuously. This research will use sophisticated machine-learning technologies, as well as predictive analytics methods, to enable the detection of deterioration in performance; to enable the prediction of possible component failures prior to occurrence. The combination of predictive analytics and advanced machine learning models will enable users and service providers to efficiently perform condition-based maintenance and make informed decisions.

This technology aims to create a more reliable vehicle system that lowers vehicle repair costs and increases safety while extending the life of an Electric Vehicle (EV). As well as meeting the objective of supporting sustainable transport, improving the consumer's confidence in EV technology and allowing EV suppliers and fleet managers to optimize their service/maintenance through data-driven decision making.

Keywords: Electric Vehicles; EV Supplier, Fleet Manager, Predictive Maintenance, Machine Learning, Post Sale Performance Intelligence.

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LIST OF ABBREVIATIONS

Table 1 List of Abbreviations

Abbreviation	Full form
AI	Artificial Intelligence
API	Application Programming Interface
BMS	Battery Management System
CNN	Convolutional Neural Network
DB	Database
DL	Deep Learning
EV	Electric Vehicle
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
KPI	Key Performance Indicator
LSTM	Long Short-Term Memory
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
OBD-II	On-Board Diagnostics II
REST	Representational State Transfer
RMSE	Root Mean Square Error
SoC	State of Charge
SoH	State of Health
UI	User Interface
UX	User Experience
V2G	Vehicle-to-Grid

1.INTRODUCTION

The global shift toward sustainable means of transportation is exemplified by the rise of Electric Vehicles (EVs) as a key part of this transition because of their capability to mitigate greenhouse gas emissions, while decreasing dependency on fossil fuels. With increasing EV adoption on a global scale, the reliability, safety and long-term performance of the electric vehicle components present a significant challenge to manufacturers, service providers, and users. Although current modern EV platforms have built-in monitoring systems that display important operational data (i.e., battery state of charge, energy consumption, and battery charging status), the monitoring functions offer primarily real-time operational data with a limited ability to diagnose conditions which are important in assessing long-term performance and determining failures of the system prior to occurrence [1], [2].

In the last few years, the world of technology has been changed completely because of new innovations in AI (artificial intelligence) and Machine Learning (ML). Intelligent prediction systems have come about through these recent innovations, which use AI to analyze large amounts of data created from the operational and sensor data of EVs (electric vehicles). Supervised learning models, Neural Networks, and Ensemble Algorithms – examples of AI related techniques (known for their ability to provide accurate predictions) have been successfully used to predict battery life, identify and diagnose fault conditions, and forecast component degradation rates of EVs. As a result, the use of these technologies fits nicely into predictive maintenance strategies for vehicle owners and service providers to deal with situations that could potentially lead to critical failure or expensive repairs prior to occurrence (2,3).

While advances in technology have allowed for the development of a wide variety of EV management platforms, many currently available solutions focus largely on pre-sale specifications, basic monitoring functions, and minimal post-sale performance intelligence and failure prediction support. The resulting research gap is created by the lack of real-time operational data available for generating useful insights about vehicle health, driver behaviors and trends related to component performance after vehicle deployment. [3], [4].

To overcome this lack of knowledge, this study will create a post-sale performance intelligence and failure prediction system for electric vehicles, as part of an AI-based electric vehicle management system. The proposed component will help electric vehicle users and service

providers through predictive analytics and intelligent monitoring systems to analyze operational data, including battery health indicators, driving patterns, charging cycles, and environmental conditions using machine learning models to identify performance deterioration, and forecast possible failures. An electric vehicle will enhance the system's reliability, reduce maintenance costs, and enable data-based decision-making processes for electric vehicle users and service providers.

1.1 Background and Context

Globally, electric vehicles (EVs) will be increasingly utilized for sustainable transportation because they can reduce CO_2 emissions and reliance on fossil fuels. Unfortunately, EV users (including fleet operators, their manufacturers, and individual consumers alike) still face performance and reliability issues related to EV battery systems because of the effect battery health has on driving range, safety, and total cost of ownership. Conventional battery management systems (BMS) provide only basic operational information, such as voltage, current, and charging status; therefore, they cannot utilize advanced analytical capabilities to analyze and interpret complicated operational data or to preemptively identify potential failures. Consequently, this inability results in a decrease in the level of proactive maintenance, which may ultimately cause unplanned failure, increased costs, and subsequently a lack of confidence in the EV market.

The use of Artificial Intelligence (AI) and Machine Learning (ML) for more efficient battery performance monitoring and predictive maintenance is currently being evaluated by researchers. For example, enhanced random forest algorithms have shown to be very effective in analyzing current battery health conditions and identifying anomalies in real-time which will enable companies to take pre-emptive action before the battery's performance deteriorates leading to a longer battery life or to mitigate operational risk [1]. In addition, new ML based frameworks for estimating battery state of charge (SoC) and state of health (SoH) were demonstrated to be more accurate than current methods because they account for the non-linear behaviors of lithium -ion batteries with regard to real-world operating environment conditions [2][3]. Furthermore, studies have concluded that using prediction modelling for estimating SoH in battery longevity as well as predicting battery failures within electric vehicles EV systems [4].

Various advancements have been made in electric vehicle (EV) technology; However, many EV management systems do not provide the means to connect Internet of Things (IoT) based data collection and processing, predictive machine learning, and analytical programming all in one platform. The purpose of this study is to propose an artificial intelligence (AI) based EV management system that uses real-time operational information collected from IoT sensors via cloud computing and includes predictive machine learning to create actionable insight into battery performance, detection of anomalies, and prediction of potential failures. The proposed system aims to improve the performance of EVs, increase reliability and support proactive maintenance with EV stakeholders.

1.2 Literature Review

Using Artificial Intelligence (AI) and Machine Learning (ML) are quickly evolving forms of research in the field of electric vehicle (EV) management, as well as battery performance intelligence. There is plentiful literature now indicating the need for predictive analytics that may facilitate proactive maintenance and assist in enhancing EV performance through means other than just their secondary applications of basic monitoring.

Many studies exist regarding predictive maintenance of electric vehicle battery systems utilizing machine learning approaches. In one example provided by Wang, he used deep learning on larger datasets for estimating the state of health (SoH) of batteries. He found that deep learning provided more accurate results than traditional regression approaches because of being able to model the nonlinear degradation behavior better than traditional regression methods [1]. In another example provided by Li, he studied using enhanced Random Forest algorithms to analyze real-time battery health data so that he could detect anomalies in battery behavior and enable maintenance before a critical failure occurs [2]. These two studies illustrate the advantages of using ensemble learning and nonlinear modelling approaches to analyzing battery degradation under different operational environments.

Artificial intelligence (AI) provides superior state-of-charge (SoC) estimation compared with traditional empirical models (chemistry and statistical models). For example, Ahmad et al. showed that neural network-based estimates of states of health (SOH) and SoC are less sensitive to noise and variability in operation of lithium-ion batteries, both of which are extremely important in actual vehicle driving conditions [3]. This work has continued to advance, as illustrated by Shao et al., who collected real-time (IoT) sensor data during battery operation and

used this in conjunction with deep learning to continuously update performance predictions [4]. Thus, AI integration will lead to improved accuracy and reliability of health estimates.

Various EV management platforms have been detailed in the literature. For instance, the open-source EV simulation framework, Battery Data Genome Project, has an abundance of data yet lacks real-time predictive maintenance through AI analytics [5]. Commercial EV management systems like onboard diagnostics from Tesla, as well as Nissan's EV Power App, provide some level of monitoring (e.g. charge and performance statistics) but generally do not have sophisticated prediction capabilities using machine learning [6]. On the other hand, research tools like EV-BAT can provide assistance with designing and testing batteries; however, there is no integrated operational intelligence between post-sale battery designs and their actual performance during real world usage [7].

Many comments have been made about the limitations of current systems. There are two main areas of concern: (1) several solutions operate using historical data sets from previous instances or only have alerts using a basic threshold so they don't have the ability to learn on a continuous basis in order to adjust to changes in driving that are unique to each vehicle [8]; and (2) few manufacturers that produce commercial products, integrate multiple sources of data (IoT sensors, OBDs, external environment) into one comprehensive assessment of a vehicle's performance [1],[4]. Lastly, due to the lack of testing of most research prototypes on large databases and/or only certain types of batteries, scaling from one type of electric vehicle to all other types of electric vehicles has not been studied, resulting in reduced applicability to mixed fleets of electric vehicles.[2][5].

The literature reviewed in this study supports the conclusion that it is possible to improve EV Battery Performance Knowledge with AI and ML. However, there is currently a significant gap in research relating to the creation of integrated and scalable post-sale EV management systems that use IoT Data Acquisition, Cloud Analytics, and Predictive Machine Learning in an integrated decision-support system. The proposed research will address these deficiencies by designing an effective and scalable Post-Sale EV Performance Knowledge and Failure Prediction System, which could generate actionable intelligence to provide stakeholders across multiple EV platforms.

Table 2 Existing research vs Proposed research Comparison

Aspect	Existing Research	Proposed Research
Data Type	Mostly battery-only metrics	Multi-modal (battery + driving + environment + charging behavior)
Analysis Method	Retrospective, offline	Real-time predictive analytics
Temporal Focus	Limited (voltage, SoC over time)	Rich temporal + behavioral patterns
Generalization	Model-specific, chemistry-specific	Cross-platform, adaptable to multiple EVs
User Application	Research prototypes, no real-world deployment	Mobile/web dashboard with predictive alerts
Intervention	Only prediction	Early intervention & actionable recommendations

1.3 Research Gap

There are numerous aspects of EV management systems utilizing AI, reflecting EV industry changes, including battery health monitoring, preventative maintenance, optimizing energy inputs, and evaluating EV performance. With continued technological advancement, further gaps exist on both an operational and theoretical basis. As such, there will continue to be a need to create larger scale, all-inclusive and real-time solutions that address the need for post-purchase performance analysis and the prediction of failure.

1. **Limited Real-Time Predictive Capabilities**

Recent studies have depended on historical offline datasets from laboratory testing or archival vehicle telemetry data. While those studies can accurately model the long-term deterioration of batteries and the progression of batteries State of Charge (SoC) in the past; these prior works have no capacity for real-time prediction to support urgent maintenance decisions in the present. When there is no way to analyze data in real-time; making any predictions about future battery

failure or optimizing the utilization of energy under an existing operational environment is not feasible, restricting the practical usefulness of models in the context of real EV deployments [1], [2].

2. **Lack of Multi-Modal Data Integration**

The current electric vehicle analytic models primarily analyze batteries and their metrics of voltage, current and temperature, thereby leaving behind many other important sources of data. For example, many operational characteristics like driving style, weather conditions, knowledge of the driver's previous charging habits as well as how much power was lost at the points of charging are often excluded from predictive models. As a result, the failure prediction performance optimization of batteries is often inaccurate or unreliable based on an inaccurate or unreliable predictive model, because the behavior and degradation of batteries is influenced by multiple sources of contextual data [3].

3. **Inadequate Temporal and Behavioral Analysis**

Multiple time-series models have been developed to analyze the performance of batteries, including the Long Short-Term Memory (LSTM) neural network model and the ARIMA time series model. These models, however, typically only model a single variable such as the state of charge (SoC) or voltage over time. There have been few studies, if any, to date that have performed integrated temporal analyses that consider multiple variables (e.g., variations in driving patterns, charging and discharging patterns, temperature fluctuations, and frequency of usage) together as the major goal of these studies was to evaluate battery performance for a specific variable without considering the impact of the other variables [4], [5].

4. **Generalization & Platform Adaptability Issues**

Existing studies have developed a number of artificial intelligence and machine learning models that rely upon a limited dataset, battery chemistry, and/or class of vehicles. As a result, many of these models operate appropriately under very limited conditions; however, they are unable to generalize across a broad range of electric vehicle platforms, manufacturers, and use cases. Further, these models do not typically account for changes that might occur as battery technologies evolve or operational conditions change, thereby limiting their practical implementation and scalability [1] [5].

5. **Absence of User-Centric Applications**

There are hardly any practical examples of the ways in which we can implement academic research for the purpose of developing systems for the general public. The absence of any system that provides real-time alerts for forecasted maintenance, predictive maintenance tips, thermal savings tips, and integration with a mobile/web dashboard creates a lack of connection with the improvements that have occurred with algorithms, and the practical use of those improvements to EV owners, fleets, and service providers [3],[6].

Novelty of Proposed Work

The proposed Post-Sale Electric Vehicle Performance Intelligence and Failure Prediction System directly address the identified voids. The primary innovations utilized in this application are:

- A real-time multimodal prediction framework that provides an integrated view of battery metrics, driving patterns, environmental variables, and charge history.
- An analysis of time-based and behavior-based degradation to help identify more complex pattern recognition over time.
- Models that can be used for all electric vehicles, regardless of type or battery chemistry.
- User-centric dashboards/applications offering predictive alerts, recommended maintenance schedules and processes, and energy efficiency best practices.

Research Gap Table

- ✓ (Tick Mark) → Means that the study addresses / includes that aspect.
- ✗ (Cross Mark) → Means that the study does not address / misses that aspect.
- Partial → Means the study partially considers it, but not fully

Table 3 Research gap table

Studies	Focus Area	Real-Time Prediction	Temporal & Behavioral Analysis	Multi-Modal Integration	AI/ML Integration	User Application	Platform Adaptability
Study 1: Traditional BMS Monitoring [1]	Voltage/ SoC diagnosis	✗	✗	✗	✗	✗	✗
Study 2: ML-based SoC Estimation [4]	Predictive SoC	✗	✓	✗	✓	✗	✗
Study 3: Deep Learning for Battery Health [2]	LSTM on lab data	✗	✓	✗	✓	✗	Partial
Study 4: IoT-enabled Battery Monitoring [3]	Multi-source monitoring	✗	✓	✓	Partial	Partial	Partial

Study 5: Predictive Fleet Analytics Models [5]	RUL prediction	✗	✓	✗	✓	✗	✗
Proposed Approach	AI & ML EV Management System	✓	✓	✓	✓	✓	✓

2. OBJECTIVE

2 Main Objectives

To design an AI/ML-driven system for real-time EV battery monitoring, predictive failure analysis, and optimization of post-sale vehicle performance.

2.2 Specific Objectives

The study has the following objectives:

- Review literature for measurable indicators of EV Battery degradation; time frame: 3 months.
- Set up IoT based Data Collection Pipeline to gather real time operational data on EVs (battery state of charge and state of health, driving patterns, charging behavior, and environmental factors); time frame: 6 months.
- Construct predictive models using ML and DL algorithms to analyze multimodal data from EVs and to predict battery performance, maintenance needs and potential failure events; time frame: 6 months.
- Assess the predictive accuracy of the models developed through the use of publicly accessible datasets, comparing them to the conventional Battery Management Systems and Rule Based Monitoring (RBM); comparison will demonstrate predictive assessment improvement.

- Validate the predictive framework in real world EV conditions during a 3-month pilot to demonstrate ability to scale, respond, and provide accurate proactive maintenance and optimization of battery performance.
- Deliver useful outputs through the use of user-friendly Dashboard or Mobile/Web Application for EV owners, fleets and service providers to provide predictive insights to assist them in Energy Efficient Operations.

3. METHODOLOGY

The Electric Vehicle (EV) Management System is being developed using Design Science and Experimental research approaches. Through this Research Methodology, predictive modeling, system implementation and verification will be accomplished through Data Acquisition which will allow for real-time out-of-the-box performance analysis as well as predictive failure detection for Electric Vehicles post-sale. Therefore, both theory and practice will be incorporated into the research method.

3.1 Research Approach

For this research project, the researcher will be following a Design Science Research (DSR) methodology, and will focus on developing and accessing a new, innovative framework for AI-based intelligence on EV performance and failure prediction after sale. Design Science is appropriate because to carry out the research and build a usable product a predictive management system for EVs that incorporates multi-modal data, machine learning models and user-centric dashboards, the research must engage in an iterative process of developing the prototype and then testing the developed prototype in a real-world environment.

3.2 Technical Methods and Algorithms

Machine Learning (ML) algorithms and Deep Learning (DL) algorithms will be leveraged to forecast battery performance and predict possible failures with the new system.

- **Supervised Learning:** SoC will be predicted using Random Forest, Gradient Boosting Machine (GBM), or Support Vector Regression while determining SoH the same way.
- **Deep Learning Model:** Long-Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) will be used to analyze battery degradation from historical driving style, charge cycles, and ambient temperature data using time-series techniques.

- **Multi-Modal Data Fusion:** To improve prediction accuracy, battery metrics, ambient conditions, and driving styles will be integrated at a feature level.
- **Anomaly Detection:** Unusual battery performance or degradation will be flagged for further investigation via the use of Autoencoders and an ensemble paradigm.

3.3 Tools and Platforms

The following tools/platforms will be utilized to develop the proposed system:

- **Programming Language:** Python is the main programming language being established as it has a large pool of available libraries for Machine Learning/Deep Learning (e.g., TensorFlow, PyTorch, and scikit-learn) and its ability to handle/organize data.
- **Internet of Things Platforms:** Controlled by either an Arduino or Raspberry Pi microcontroller that provide sensors to gather real-time data with respect to the operations of the electric vehicle.
- **Cloud Platform:** The AWS IoT platform or the Microsoft Azure IoT Hub will serve as the secure storage location for the data; additionally, AWS IoT and Microsoft Azure IoT provide real-time analytics and deploy machine learning models.
- **Visualization Tools:** Dash or Streamlit will allow the user to view both the user-friendly dashboard and actionable insights.

Reason: These four tools will enable the processing of real-time data across several different data sources for developing and deploying machine learning models, and this capability will allow for building a scalable predictive analytics system.

3.4 Data Collection and Ethical Considerations

- **Data Sources:** Telemetry live streaming from Electric vehicle IOS sensors. Also, a number of publicly available datasets on EV's such as how fast batteries deplete or how quickly it charges and a myriad of environmental data.
- **Ethical considerations:** The work performed in this study is strictly compliant with all Institutional Ethical Guidelines and Regulations that govern how employees handle operational and personally identifiable information on their vehicles. All participants' data will be anonymized to protect the user's privacy.

3.5 Validation Strategy and Performance Metrics

The method for measuring how successful the prediction model is includes:

- Model Validation – Cross-validation, testing on unseen data, comparing against baseline models (e.g., standard BMS and rule-based monitoring),
- Performance Metrics – Regression metrics for accuracy in predicting State of Charge (SoC) and State of Health (SoH): Root Mean Square Error (RMSE), Mean Absolute Error (MAE) as regression metrics; the use of classification metrics (e.g., accuracy, F1 Score, precision (true positives vs. number of samples predicted as positive), recall (true positives vs. number of actual samples)) to measure how accurate we are at predicting battery failures (predicting when a battery will fail) and detecting anomalies.
- Real-world testing – the use of selected EVs to monitor the operational performance of the predictive model in real time to assess the scalability, speed, and usability of the tool.
- User feedback – provide a means to collect user inputs/feedback from EV owners/fleet operators/service providers on the effectiveness of their dashboards and actionable recommendations provided by the prediction model.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

This system leverages artificial intelligence to create an intelligent EV performance monitoring and failure detection framework that will provide real-time information to both consumers and fleet operators about the condition of their electric vehicles, by utilizing real-time vehicle data acquired via IoT technology, analyzing that data using machine learning and deep learning techniques, storing that data in the cloud, and delivering usable dashboards to end users.

The entire solution is designed to be:

- Cross-Platform (works on multiple hardware and software systems)
- Scalable (grows with demand)
- Secure

The entire solution used to create this intelligent EV Performance/Intelligence framework will be composed of four main elements for collecting data and delivering real-time information:

- **IoT Data Acquisition Layer:** Collects real-time operational data from the electric vehicles using various IoT sensor networks.
- **Cloud Storage and Preprocessing Layer:** The collected data will be processed in the cloud and prepared for Artificial Intelligence/Machine Learning use.
- **AI/ML Analytics Engine:** Will analyze the stored data and build predictive models that estimate the State of Charge (SoC) and State of Health (SoH) of the battery and analyze data for anomalies and future potential failure.
- **User Interface & Dashboard Layer:** The end users (EV owners, Fleet Operators, and Service Providers) will have access to interactive dashboards that provide actionable information and predictive alerts.

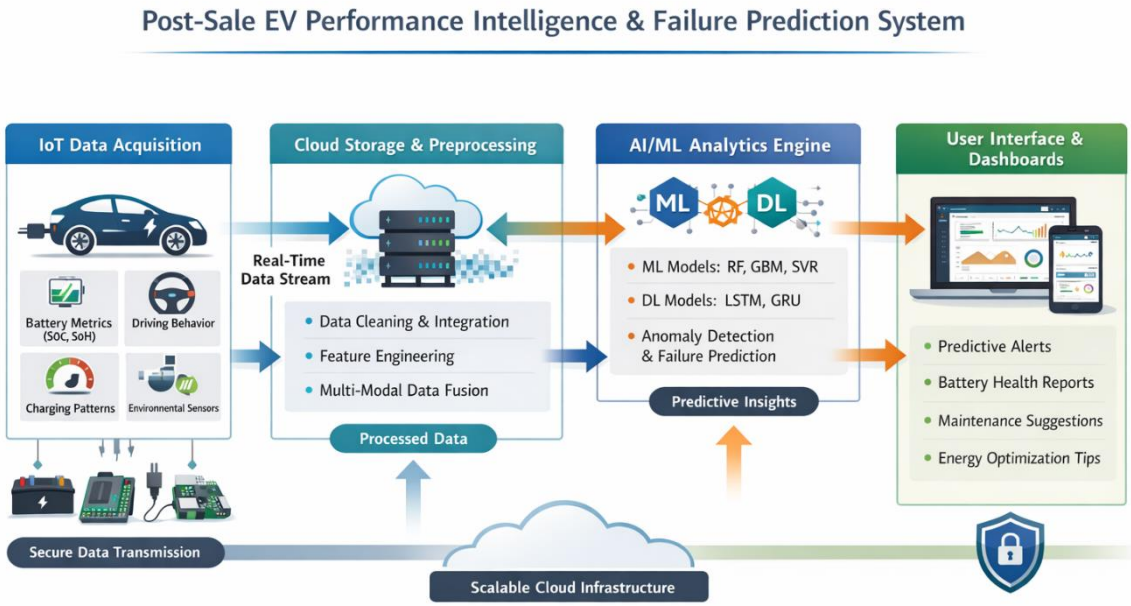


Figure 1System Architecture

4.2 Individual Component Description

IoT Data Acquisition Layer

- Capture EV metrics (e.g., battery SoC, SoH, driving patterns, charging event, environmental conditions) using sensors and microcontrollers (i.e., Arduino and Raspberry Pi).
- Streaming the collected data to the Cloud securely using either MQTT or HTTPS for Real-Time Analytics.

Cloud Storage and Preprocessing Layer

- Validation, filtering, normalization, and feature extraction of data.
- Integrating site-specific, multi-modal data streams into structured datasets for modeling uses.
- Providing secure storage, access control, encryption and backups.

AI/ML Analytics Engine:

- Generate Battery Health, Failure Prediction, and Anomaly Detection with Predictive Models (e.g., Random Forest, Gradient Boosting, LSTM/GRU).
- Generate Recommendations to Optimally Operate and Provide Data Insight as Opportunities for When to Make Maintenance Repairs.
- Continuously Update Models using Real-Time Data as it Becomes Available (to support Continuous Adaptive Learning).

User Interface & Dashboard Layer

- consists of web and mobile Dashboards that allow users to see alerts for predictive maintenance, health of battery, and how to maintain their vehicle.
- The user interfaces will include ability to interact with the system, receive notifications, and see trends and forecasting (historical data).
- Additionally, the Dashboard provides APIs for interaction with other fleet management software and service providers.

4.3 Component Interactions

- Data Acquisition: Performance and operational data from the electric vehicle are captured in real-time by onboard IoT sensors. This data includes battery metrics (state of charge, state of health), driving behavior, charging patterns, and environmental conditions.
- Data Transmission: The collected data is then sent securely to the cloud infrastructure over the network via the IoT Platform with minimal latency and encryption for privacy and security.
- Data Processing & Storage: Incoming data is pre-processed in the cloud (cleaning, normalizing, feature engineering) and fused using multi-modal data. The processed data set will be stored in both immediate and historical formats to facilitate analytics.
- AI/ML analytics engine: The preprocessed datasets go into the ML and DL models for detecting an anomaly, predicting if there will be a battery issue, assessing the battery's health, and optimizing battery performance based upon a variety of sources of data collected over time.
- User Interaction/Feedback: The results of predictions and actionable insights are delivered back to the end users (EV owners, fleet managers, service providers) via dashboards and mobile/web user interfaces. The system also generates predictive alerts, maintenance recommendations, and energy optimization suggestions.
- Continuous Monitoring/Support: This architecture supports the ongoing monitoring of the performance of EVs enabling real-time decision making, proactive maintenance, and increased operational reliability.

4.4 Data Flow Explanation

This paper discusses a systematic flow of data in a proposed system that includes real-time monitoring, predictive analysis, and useful information about Electric Vehicles (EV) performance; the following is an outline of how this would work.

1. Data Generation
    - The EV sensors enabled with IoT generate telemetry data at a constant rate that measure Battery Values like (SoC, SoH), Driving patterns, Charging patterns, and Environmental conditions.

2. Data Transmission
    - Telemetry data transmitted from IoT devices (i.e. the EV sensors) to the cloud is via an MQTT/HTTP connection. This means that low latency and encryption are assured between the IoT devices and the cloud.
3. Data Preprocessing
    - Telemetry data captured from IoT devices undergoes cleansing, normalizing, and restructuring to provide structured, analyzable datasets to users' applications.
    - The data from the aforementioned datasets are fused together to provide greater predictive capability by bringing together battery, driving behavior, and environmental conditions datasets into one dataset that will have enhanced algorithms.
4. Predictive Analytical Modeling

- Machine-learning (ML) methods (e.g., Random Forest, Gradient Boosting, SVR) and deep-learning (DL) methods (e.g. LSTM, GRU) were used to analyze preprocessed datasets.
- The system will analyze patterns and trends and predict when batteries will degrade, when they will fail, and when there is an anomaly.

5. Visualization and Alerting

- Users will receive insight into battery status through an interactive dashboard that includes predictive alerts, battery state of health report, maintenance recommendations, and energy optimization.
- Alerts will notify users of adverse conditions and/or when a failure is imminent to enable timely repair or replacement.

Outcome:

Utilizing this comprehensive data flow, real-time performance assessment of EVs, predictive failure detection, and data-driven decisions will all help improve battery longevity, promote safe operation, and increase consumer confidence in EVs.

4.5 Scalability and Security Considerations

Scalability

This design supports the ability to grow the amount and types of data generated by Electric Vehicles (EVs) and the number of EVs connected to the system without corrupting system functionality.

- Modular Architecture: The overall design supports independent functionality of each layer of the system; therefore, adding more EVs, types of EV batteries and/or sources of data does not affect any of the system's existing functional components.
- Cloud-based Processing: A scalable cloud infrastructure allows for real-time monitoring of large fleets and physical distribution of EVs through elastic scaling of storage, compute and analytical resources.
- Model Adaptability: Continuous predictive accuracy can be maintained by using AI/ML models that can be retrained and updated incrementally in order to adapt to new battery chemistries, driving behaviors or environmental factors.
- Multi-Tenant Support: The system supports multiple users at one time by providing each user with a unique dashboard and data stream, allowing for personalized insights.

Security Considerations

The protection of data and privacy is paramount because of the nature of EV (electric vehicle) operations, as well as EV user data.

- Data Encryption: All telemetry data that is transmitted from IoT devices to the cloud is sent securely through an end-to-end (TLS/SSL) data encryption form of communication. All stored data is encrypted using AES-256, or equivalent data encryption standard.
- Access Control: Role based access control provides controls for the EV user to manage their access to certain segments of data (e.g vehicle owner, fleet manager, etc.).
- Secure Cloud Infrastructure: The managed cloud servers are protected by multiple security measures, including firewalls, intrusion detection systems, and some regular security audits.

- Data Anonymization: In an effort to protect EV user's privacy and other personally identifiable information, PII will be modified by the data processor or protected through storage prior to processing for long-term storage.
- Fault Tolerance & Redundancy: Backup servers and failover mechanisms ensure the availability and reliability of the EV Cloud Service in the event of hardware or network failure.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholders involved or benefited by the proposed system providing performance intelligence for Electric Vehicles (EVs) and predictive failures include:

1. EV Owners

users who will need information and updates regarding battery condition, how to improve battery life, and advisory notices on how to lessen the chance of battery failures (unexpected or unexpected).

2. Fleet Managers

people in charge of managing many vehicles (EVs) and need to pull together reports to monitor the performance of EVs, conduct preventive maintenance, and reduce operating costs.

3. EV Service & Maintenance Providers

automotive suppliers and manufacturers providing products and services needed to prevent and repair failure and provide data to help with maintenance timing.

4. EV Researchers

people/groups that need to obtain EV performance data that would be kept confidential and used to develop better safety standards, improve energy use, and support the process of using EVs.

5.2 Functional Requirements

- Real-Time Data Capture: The system will gather telemetry data from EV sensors, including data about the battery's performance, how the driver drives the vehicle, how the battery is charged, and how the environment affects the battery.
- Data Storage and Management: The system will securely store and categorize the structured data gathered from the performance of EVs for historical and real-time analysis.
- Predictive Analytics: The system will use A.I./M.L.-based predictive techniques to predict when a battery will have issues, when it will fail, and when an opportunity exists to maximize energy efficiency.
- Anomaly Detection: The system will identify unusual trends in the performance of an EV's battery and in EV operations.
- Dash Board Visualization: The system will provide users with visualizations of EV performance through interactive dashboards.
- User Alerts and Notifications: The system will alert users to possible failures, but advise for maintenance or that their EV has not been performing as expected using dashboards, mobile applications, or e-mail.
- Multi-Vehicle and Multi-Platform Capability: The system will support using multiple EVs or similarities in EVs with the same battery type and regular use in the same environment.
- Data Export and Reporting: The system will allow users to export fleet management reports to provide maintenance schedules and to comply with applicable law.

5.3 Non-Functional Requirements

- **Security:** Data transmission and data storage will be encrypted. Data in-transit will be secured using TLS/SSL, while stored data would use AES-256 encrypted format. Only users with role-based permissions will be able to access and use these securely encrypted data.
- **Performance:** System will provide real-time processing and analysis of data streams from multiple EVs (Electric Vehicles), or in other words, there should not be any noticeable latency between when data is received from EVs and when it is processed or analyzed.

- **Scalability:** The system must be designed to horizontally scale to meet the growing number of EVs, volume of data, and number of concurrent users.
- **Reliability & Availability:** The system shall provide reliability and availability by being able to operate continuously with minimal downtime through use of a fault tolerant cloud infrastructure with redundant servers.
- **Usability:** The dashboards and mobile interfaces must be designed to be intuitive, visually clear, and actionable by both technical and non-technical users.
- **Maintainability:** The system shall be easily maintainable with the ability for incremental updates to AI/ML models, dashboards, and IoT integration without interruption to existing business operations.
- **Compliance:** The system shall comply with all applicable data protection regulations, including anonymous user identification and anonymizing user identifiable information.

5.4 Use Case Scenario

Use Case Scenario

Table 4 Use case Scenario

Number	1
Name	Monitor EV Battery Health & Predictive Failure
Summary	The system provides ongoing monitoring of EV battery metrics (SoC, SoH, voltage, temperature), driving habits, and environmental conditions in real-time. The system produces an alert when a discrepancy or potential problem appears in the data collected. Dashboards or mobile apps provide the EV owner with forecasts of future performance as well as insights into how they drive their vehicle.
Priority	5
Preconditions	• IoT-enabled sensors and/or OBD-II have been installed into the EV. • Ability to send telemetric data to the

	cloud is operational.	
Postconditions	• Real-time battery health scorecards and trends are shown on a dashboard for users. • Users will receive predictive alerts for maintenance recommendations. • Performance and telemetry data are logged for historical trend analysis to support model retraining.	
Primary Actor(s)	EV Owner	
Trigger	New telemetry data is generated from the EV.	
Main Scenario	**Step**	**Action**
	1	EV sensors collect battery and operational information (e.g. state of charge (SOC), state of health (SOH), temperature, voltage, driving behavior, and environmental conditions).
	2	The telemetry data is received through the Internet of Things (IoT) platform.
	3	Once the data is received, it is preprocessed, cleaned, and normalized.
	4	Features such as metrics (i.e. battery), trends (i.e. driving behavior), and environmental variables are extracted from the processed data.
	5	The predictive models created using machine learning (ML) and deep learning (DL) methods will then be used to analyze this data to provide battery performance forecasts as well as predict possible future failures.
	6	The results of the predictive analysis will be delivered through a dashboard and/or mobile

		application for the user to view battery performance, health statistics, and forecasted alerts.
	7	The user will receive alerts and/or recommendations if the data has exceeded pre-established thresholds or the risk of an impending failure has been determined.
	8	The data will also be stored in the cloud for further (i.e. future) model retraining and long-term (i.e. trend) analysis.

Extensions	**Step**	**Branching Action**
	2.a	If the telemetry connection does not succeed, the system will resend the telemetry data and provide the user with an anticipated delay for completion of that transmission.
	4.a	If the telemetry data is corrupt or does not exist the system will mark the data as incomplete and rely on historical analysis or estimate of the data.
	5.a	If the model's confidence is low the system will schedule additional collection of data or notify a user that the model's confidence is low.
Open issues	1	Facilitating predictive cutoffs for battery alerts and battery failure identification.
	2	Developing means of supporting the variety of battery chemistries and specifications present in electric vehicle (EV) models.
	3	Securing the transmission of data and preserving user privacy by securing the handling of multi-source telemetry.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

- **Primary Market:** Individual owners of EVs who require predictive maintenance, reliability, and optimization of energy use.
- **Secondary Market:** Fleet operators of multiple Electrically Powered vehicles (EPVs) (e.g., logistics companies, rideshare companies, and delivery firms).
- **Categories for market or customers:**
    - **Electric Vehicle (EV) Owner** – Technology and commutability are important. EV owners between 25 years and 45 years old, with personal use of individual EVs, want dashboard-like insights on how their EV uses energy and predictive alerts for maximum battery life and minimum maintenance cost.
    - **Electric Vehicle Fleet Management** – Need to manage between 10 and 500 Electric Vehicles. Fleet managers between 30 and 55 years old want to receive real-time analytic reports on the performance of the vehicles of their electric fleet in order to provide predictive notifications of failures in order to minimize the total cost of electric vehicle operation.
    - **Electric Vehicle Manufacturer / Service Provider** – Interested in receiving actionable insights for early preventative maintenance, warranty/maintenance planning, and improving/optimizing customer satisfaction.

6.2 Value Proposition

- Provides timely analysis and forward-looking forecast of the battery's operational characteristics.
- Utilizes preventative maintenance to help prevent unexpected failures and related expenses.
- Utilizes behavioral and operational data to maximize the life and performance of batteries.
- Works with all EVs regardless of model or type of battery chemistry.
- Easy to use applications and dashboard provide clear and actionable recommendations to all users.

6.3 Revenue Model

- Subscription model: provide both monthly and annual subscriptions to the individual end user of an EV and a fleet operator.
- Enterprise license: Fleet operators and service providers pay for enterprise access to analytics dashboard and APIs.
- Freemium model: Basic battery health monitoring would be free to all users, but predictive alerts, comprehensive analytics, and trends through history would be offered on a premium basis.
- Data services & insights: EV performance data aggregated and anonymized can be purchased on a subscription basis by research institutions, energy companies, and/or manufacturers.

6.4 Cost Estimation

- Costs associated with development (including but not limited to) AI/ Machine Learning development, Internet of Things integration, Cloud infrastructure establishment.
- Hardware costs (including but not limited to) cost of Internet of Things sensors, cost of Electric Vehicle telemetry module(s) and cost of server infrastructure.
- Operational costs (including but not limited to) Cloud Storage and/or Cloud Computing costs, maintenance costs, software update costs, customer support costs.
- Marketing and Distribution costs (including but not limited to) digital marketing, app deployment, and customer onboarding.
- The estimated cost range is about Rs 25,000 to Rs 75,000 to develop the first version of the prototype system and create basic infrastructure to do so. Total costs will depend upon the features of the various systems, what type of information is required to use them (i.e., data) and how many electric vehicles have been included within the research context.

6.5 Pricing Strategy

- Individual Electric Vehicle Owners: Predictive features at LKR 300 to 10000 per month for additional functionality.
- Fleet Owners: A tiered price structure depending on how many vehicles are in a fleet will apply (e.g., for fleets of up to 50 electric vehicles, the cost is LKR 60000/month and increases as the fleet grows).

- OEMs and Service Providers: Customized pricing based on service level as well as integration and analytic services.
- Freemium Model: Basic battery monitoring is an additional free service to increase interest in battery monitoring and motivate users to upgrade.

6.6 Competitive Advantage

- Real Time Predictive Multi Modal: Unlike competitors who focus on predicting only voltage and SoC, we have implemented a hybrid of battery metrics, driving styles, charging patterns and the environment into our predictive modeling.
- Platform Independent Design: Our Software/Hardware solutions are designed to work with all makes and models of EV, and with all battery technologies.
- User Centric Dashboards: Dashboards provide actionable data to the end user(s) vs. just raw data or static reports.
- Predictive Insights & Prescriptive Insights: By providing prescriptive as well as predictive failure info, our solution provides optimization of your costs and efficiencies.

6.7 Intellectual Property Considerations

- Patent Opportunities - AI and ML model-based approaches to predicting failures in batteries across multiple modes, along with predictive maintenance frameworks.
- Copyright/Software Protection - Dashboard UI, mobile app interface, and unique algorithms used to forecast battery performance
- Data Privacy Compliance - Local and global regulations governing how user information (e.g., anonymized electric vehicle performance data) must be handled.

7.BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Table 5 Hardware Cost

Item	Quantity	Estimated Cost (LKR)	Justification
IoT sensors (voltage, temperature)	5 units	20,000	Simulate EV battery data col. for proto. test.
Edge processing unit (Raspberry Pi / microcontroller)	1 unit	8,000	Performs basic, local pre-process of sensor data.
EV OBD-II adapter	2 units	6,000	Allows connection with EV diagnostic systems.
Miscellaneous cabling & hardware	-	6,000	Required to setup sensors and provide connectivity.

Total Hardware Cost: 40,000 LKR

7.2 Software Licenses

Table 6 Software Licenses

Software	License Type	Estimated Cost (LKR)	Justification
Python libraries & ML frameworks	Open-source	0	TensorFlow, PyTorch, and Scikit-learn are free tools.
Dashboard & Visualization (open-source)	2 users	10,000	Visualization using tools like Grafana or free BI tools.
IoT Middleware / Edge software	Open-source	5,000	Data processing and device communication tools.

Total Software Cost: 15,000 LKR

7.3 Cloud Services

Table 7 Cloud Services

Service	Usage	Estimated Cost (LKR)	Justification
Cloud Storage & Database	100 GB	10,000	Stores EV telemetry and prediction results.
Cloud Compute / ML Training	Small instance	15,000	Used for model training and testing.
IoT Data Ingestion Service	Prototype usage	5,000	Handles basic data transmission from devices.

Total Cloud Services Cost: 30,000 LKR

7.4 Human Effort Cost

Table 8 Human Effort Cost

Role	Duration	Estimated Cost (LKR)	Justification
Student Developers / Research Team	Project duration	15,000	Covers minor expenses, internet, and development support.

Total Human Effort Cost: 15,000 LKR

7.5 Total Estimated Budget (LKR)

Table 9 Total budget

Category	Total Cost (LKR)
Hardware	40,000
Software	15,000
Cloud Services	30,000
Human Effort	15,000
Total Budget	**100,000**

8.WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown Chart

To ensure that the entire project is developed systematically and the allocation of tasks is effectively organized, the entire project is divided into several components and work packages. The hierarchical breakdown of the EV Intelligent Monitoring and Forecasting System into manageable tasks and subtasks is shown through the Work Breakdown Structure (WBS) in Figure 2

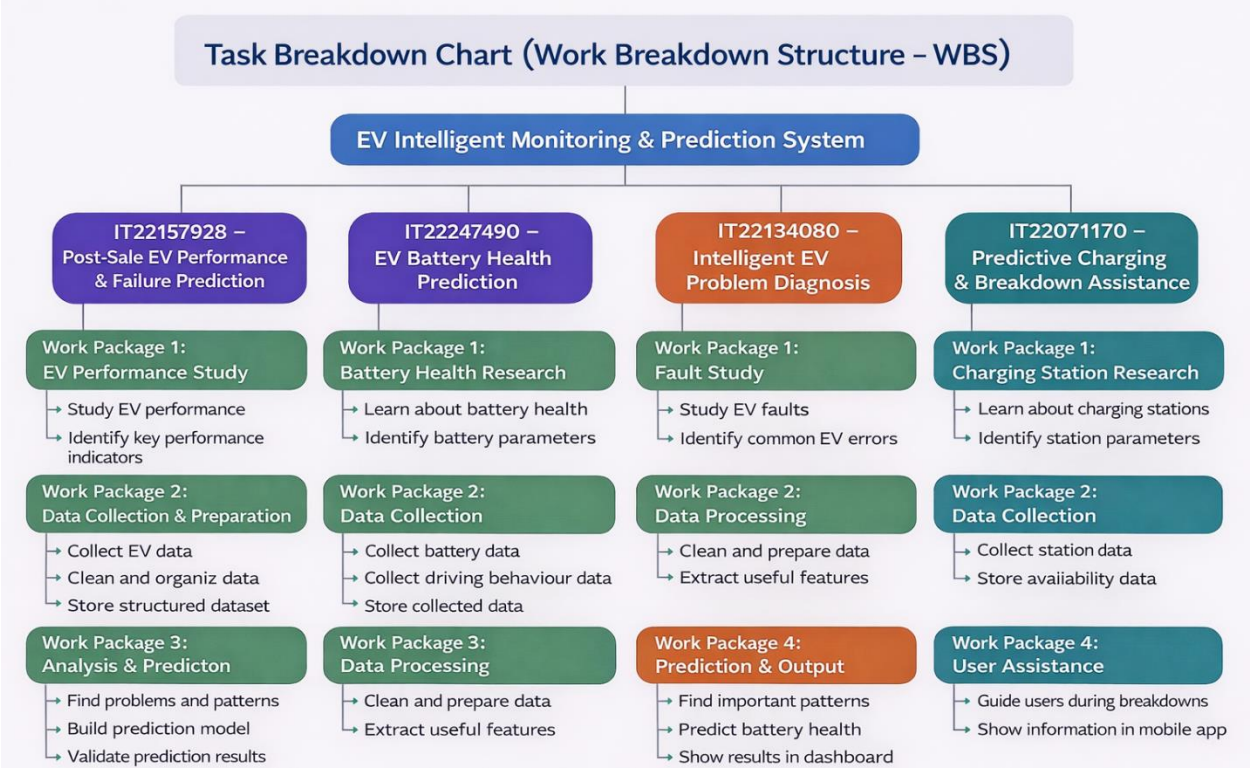


Figure 2 Work breakdown structure

As shown in Figure 8.1, the entire system is divided into four major parts, namely, aftermarket EV performance and failure prediction, EV battery health prediction, intelligent EV problem diagnosis and predictive charging, and breakdown support. The components are further divided into different work packages to ensure a clear development flow and distribution of responsibilities.

Focusing on the EV battery health prediction component, the work is divided into three main phases, namely battery health research, data collection, and data processing.

These phases include determining critical battery parameters, collecting both battery and behavioral data, and data cleaning and feature extraction. This formal model

ensures that the prediction model is created with properly prepared and meaningful information.

At the same time, the other parts have a logical flow of research and data collection for their processing, prediction, and user support. Such a hierarchical decomposition helps manage project complexity, allocate tasks, and develop all components of the system in a coordinated and systematic manner.

8.2 Balanced Workload Distribution

The table below shows the workload shared by the team members working on the project. It describes how responsibilities are assigned, work is shared, and collaboration is maintained to ensure the successful development of the system.

Table 10 Balanced workload distribution

Aspect	Description
Task Allocation	Each team member is responsible for one major system component.
Work Distribution	All members contribute equally through research, system design, model development, implementation, and testing.
Collaboration	Team members collaborate during system integration to combine all components into one complete EV Intelligent Decision Support System.
Project Outcome	The modular development approach ensures balanced workload, efficient development, and a fully integrated intelligent EV decision support platform.

As shown in Table 8.2, the project is organized in terms of systematic workload distribution. Team members will be assigned to certain parts of the system, which means that tasks and responsibilities are clearly distributed. Meanwhile, all members are engaged in key activities such as research, system design, model development, implementation, and testing, which

encourages equal contributions. Teamwork is relevant during the system integration phase, where separate components are integrated into a fully functional EV intelligent decision support system. This integrated or coordinated style helps keep the development process efficient and balances the team's workload with the result.

GANTT CHART

Figure 3 is a layout of the Research Project (RP) schedule for 2026. It depicts the significant phases of the project and important project deliverables throughout the year. It is a visual representation of the progress of the project from the initial stages to the final stages.

Task	1	2	3	4	5	6	7	8	9	10	11	12
Project Planning & Proposal Preparation	■	■										
Literature Review	■	■	■									
Requirement Analysis		■	■									
IoT Data Collection Setup			■	■								
Data Preprocessing & Storage				■	■							
Feature Engineering				■	■							
ML/DL Model Development					■	■	■					
Model Training & Optimization						■	■	■				
Dashboard & System Development							■	■	■			
System Integration								■	■			
Testing & Validation									■	■		
Pilot Deployment										■		
Final Evaluation & Improvements										■	■	
Final Report & Thesis Submission											■	■

Figure 3 Gantt chart

The project begins in January and February with planning activities such as proposal submission and presentation. Development begins in February, and this continues with other key activities such as submission of checklists and regular progress reviews. In the final phase of the project from October to December, the main activities include delivering key deliverables, including website submission, final presentation, submission of research papers, and preparation of publication evidence. The project is well planned and outlined for smooth progress and completion.

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11 APPENDICES

11.1 Survey Instruments: Post-Sale EV Performance Intelligence & Failure Prediction System

Survey Title: Survey to Evaluate Awareness of EV Battery Condition and Predictive Maintenance

Guidelines: Please complete all questions honestly; responses are anonymous and will only be used for academic purposes. It will take approximately 10 minutes to complete this survey.

Part A – Participant Profile

Q1. Your Age:

Select Age Range:

☐18-24

☐25-34

☐35-44

☐45-54

☐55+

Q2. Your Employment type:

Choose one of the following:

☐Student

☐Government Contractor

☐Private Contractor

☐Business Owner

☐Other.

Q3. How long have you owned your EV?

☐Less than 6 months

☐6-12 months

☐1-3 years

☐More than 3 years

Q4. What type of EV do you own?

Select one of the following:

☐Battery Electric Vehicle (BEV)

☐Plug-In Hybrid Electric Vehicle (PHEV)

☐Hybrid Electric Vehicle (HEV).

Part B-Charging Behavior

Q1. Asked about daily charging for your electric vehicle (EV), could you please tell us how many times you charge per week?

☐One time.

☐Two to three times.

☐Four to five times.

☐Every day.

Q2. When asked about what kind of charger you use to charge your EV, please choose either of the following options:

☐Home (level 1/level 2).

☐Public.

Q3. When asked at what percentage you generally start charging your EV, please pick your choice:

☐Less than 20%.

☐20-40%.

☐40-60%.

☐More than 60%.

Q4. When asked if you regularly charge your EV battery to full capacity, please would you select:

☐Yes.

☐Sometimes.

☐Rarely.

☐No.

Q5. When asked if you have had any unexpected deterioration in battery performance or range, please select your answer:

☐Yes.

☐No.

☐Not Sure.

Part C-Driving Habits

C1: How far do you drive every day?

☐Less than 20 km

☐20 - 50 km

☐More than 50 km

C2: How do you drive?

☐Gently

☐no acceleration without using extraordinary force (e.g., hard braking or acceleration) OK sometimes

☐ Variable depending on circumstance

C3: Do you usually drive at 100 km/hr.?

☐Yes – frequently

☐Occasionally

☐1 time per week or less.

C4: How often would you say you drive for distances shorter than 5 km?

☐All the time

☐multiple a week

☐once a month or greater.

Part D – System Acceptance & Willingness to Use AI/ML Prediction System

Q1. If a predictive battery health prediction system was available how much would you be willing to spend a month for such a service?

☐Free only

☐LKR 500 – 1000

☐LKR 1000 – 1500

☐LKR 1500 - 2500

☒Over 2500

Q2. ow important is it to you that your personal information would be kept confidential when using a predictive battery health monitoring system?

☐Extremely important

☐Very important

☐Moderately important

☐Not important

11.2 Preliminary UI Mockups

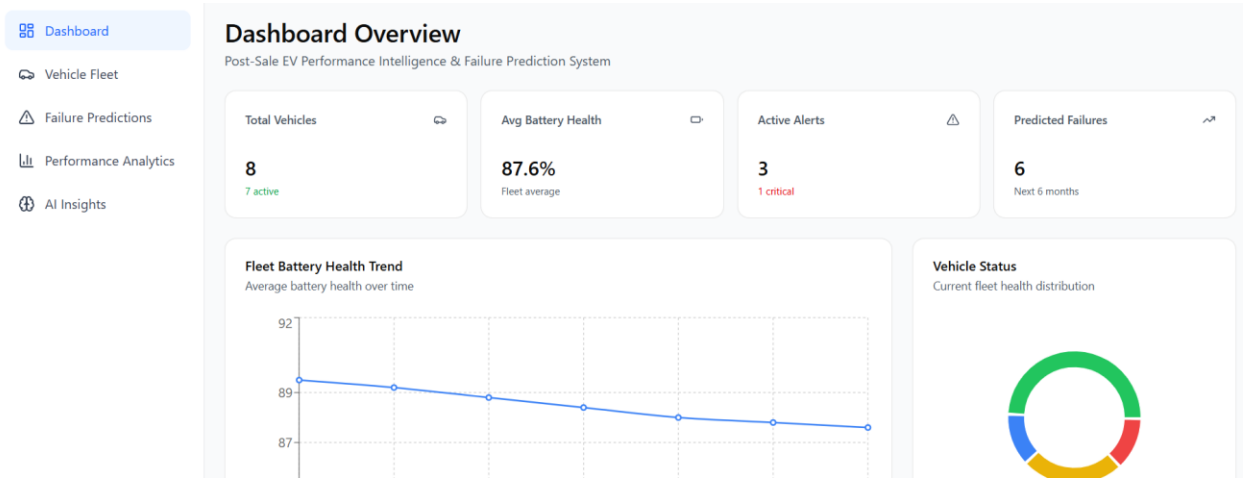


Figure 4 UI 1

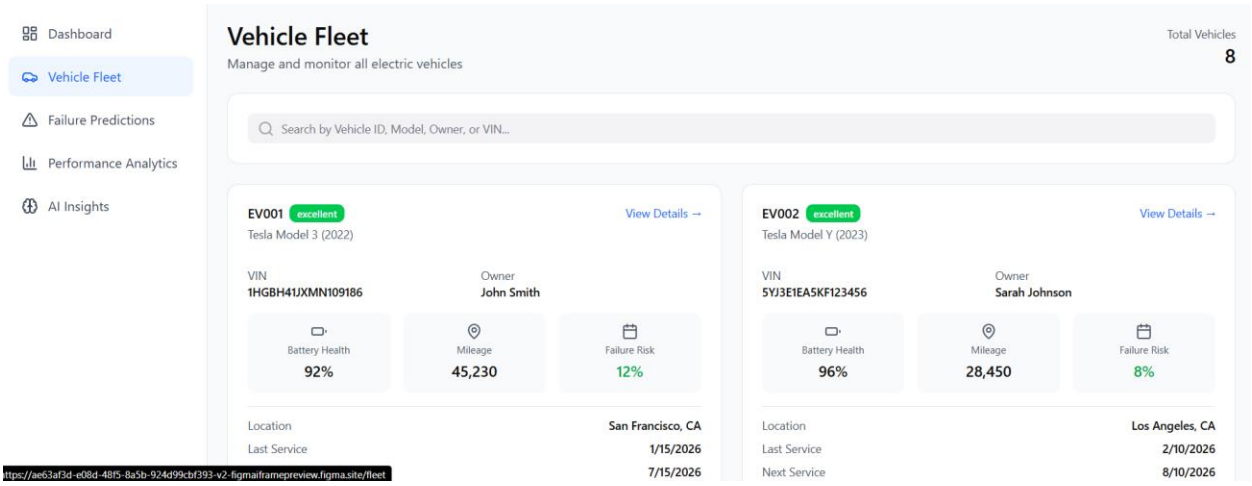


Figure 5 UI 2

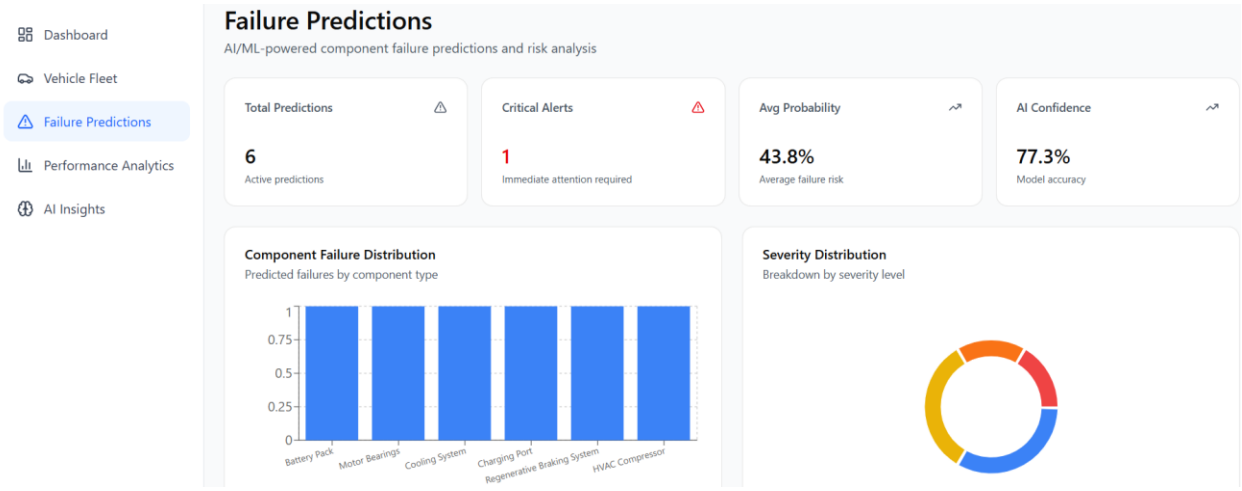


Figure 6 UI 3

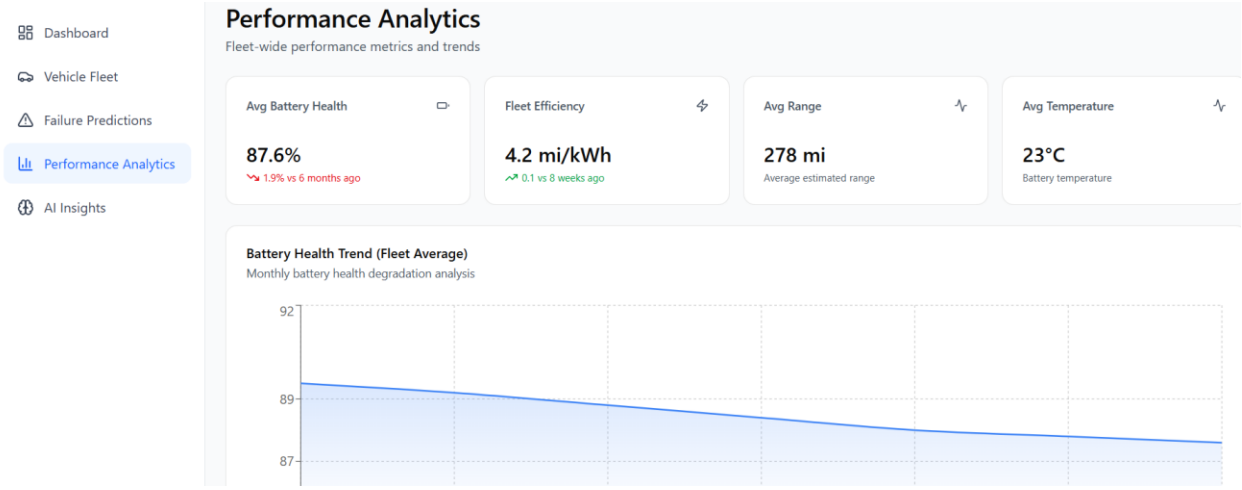


Figure 7 UI 4

11.3 Risk Analysis

Table 11 Risk Analysis

Risk	Impact	Likelihood	Mitigation / Control Measures
Sensor data loss or corruption	High – Could affect model accuracy and system reliability	Medium	Redundant sensors, Secure Data Transmission, Cloud Backup
Incorrect AI/ML predictions	High – Could mislead users, reduce trust in the system	Medium	Validate models with historical data, retrain often and monitor continuously
Unauthorized access to EV data	High – Risk of data breach, privacy violation	Low	Apply strong encryption, authentication, and role-based access control
User resistance to predictive insights	Medium – Users may ignore alerts or recommendations	Medium	Provide clear explanations of AI recommendations and user education
System downtime / service interruption	Medium – Reduces system availability for fleet managers and owners	Low	Use failover servers, cloud redundancy, and continuous system monitoring
Battery performance misinterpretation	Medium – Users may misinterpret reports and take wrong actions	Medium	Include clear explanations, thresholds, and guidance in reports and dashboards

Regulatory / compliance issues	High – EV data handling may violate local regulations	Low	Ensure compliance with data protection laws and obtain ethical approvals

11.4 Ethical Documents

Purpose: The purpose of this Appendix is to describe the ethical guidelines applied in the development and investigation of the Post Sale Electric Vehicle (EV) Performance Intelligence and Failure Prediction system which relies on the collection, storage, and processing of information on users' performance with electric vehicles (EVs) in order to develop predictive models that use artificial / machine learning to provide predictions related to battery health.

1. Informed Consent

All research participants in surveys, usability assessments, and data gathering must provide full informed consent before participating. Participants will be informed of:

- The aim of the study
- Types of data (batteries, driving, charging, distances, performance)
- How the data will be utilized & accessed
- Their option to abandon the study anytime without penalty

Informed consent must be given before collecting any data.

2. Minimizing Data

Only the minimal amount of data necessary for predicting battery health or clarifying reasons for failure will be collected.

- No sensitive personal information will be collected; e.g., financial information, identity cards, or precise GPS location(s) from the point of departure to the destination.

- Data collection is kept to a minimum in order to facilitate the operation of the system and help with the fulfilment of the associated research.

3. Protection of Privacy and Anonymization

- All PII will be anonymized.
- When PII is not used in dataset construction, only the participant's identification number will be used (and no other identifiable number).
- Whenever aggregated data is supplied to the manufacturers or service providers, it will have the individuals removed to ensure that the individuals cannot be determined from the supplied data.
- Data will not be used in training AI/ML systems in a way that allows an individual to be identified from their data.

4. Data Security

- The data gathered will be kept on secure servers that are password protected, allowing access only to those people who are authorized to conduct academic research.
- No third parties will have access to any of the research data unless written consent has been given by the participant.
- At the conclusion of the research project, all research data will be destroyed unless the participant gives permission for the continued use of the data.

5. Participation is Voluntary

- All of the research procedures, including surveys and interviews, and usability tests are completely voluntary.
- No financial compensation will be offered to influence participation.
- Participants can withdraw from any of the research procedures at any time for any reason without penalty.

6. Avoiding Causing Harm

- The research will not cause anyone to experience physical, psychological, financial, or reputational harm.

- Questions in surveys and usability tests are non-invasive and do not violate the privacy of the participants.
- For any concern or complaint that a participant has, the researcher will attempt to address the concern or complaint immediately, including the possible removal of the participant from the research study if requested.